# Idea generation & development

Higher · Design · Worksheet

D&M WITH MR KELLY

Name _____ Class _____ Date _____

**By the end of this worksheet I can…**

- ☐ describe the key stages of brainstorming
- ☐ describe the key stages of morphological analysis
- ☐ explain why idea-generation techniques are used
- ☐ describe the key stages of analogy
- ☐ describe how lifestyle and mood boards are used
- ☐ explain exploration, synthesis and refinement of ideas

## I DO — Worked example

Read how a full-mark answer is built

*A design team needs ideas for a new bike light.*

**1 Describe three key stages of brainstorming.** (3 marks)

**Full-mark answer** each ✓ is one mark

The problem is explained clearly to a small group so everyone knows the purpose ✓. The group suggests as many ideas as possible within a time limit, and every idea is recorded ✓. No idea is criticised; members build on each other's ideas before the best are selected ✓.

**Why marks are not awarded**

*"You think of lots of ideas and write them down."*

0–1 mark: "think of lots of ideas" was given 0; only recording ideas could score.

## WE DO — Guided practice

Use the starters and word bank to help you

**2 Describe two idea-generation techniques.** (4 marks)

**Sentence starters**

Brainstorming: …
Morphological analysis: …
Analogy: …
Mood board: …

**Word bank**

Morphological analysis · technique · Mood board · problem · simply · stated · brainstorming · Analogy

Tip: 4 marks = 4 separate, developed points.

**YOU DO** **On your own** No notes. Check your answers afterwards.

### 3 Quick check: tick one box for each.

**a)** Combining the best features of several initial ideas is called...
☐ Synthesis ☐ Refinement ☐ Evaluation ☐ Analysis

**b)** Which is a rule of brainstorming?
☐ Only feasible ideas are written down ☐ Every idea is recorded without criticism ☐ Ideas are judged as soon as they appear ☐ It must be done alone to avoid copying

**c)** Nike basing a trainer's cushioning on Formula One suspension is an example of...
☐ Mood board ☐ Morphological analysis ☐ Analogy ☐ Synthesis

**d)** How does morphological analysis improve the breadth of ideas?
☐ It creates many unique combinations of variables ☐ It quickly removes the weakest ideas early on ☐ It tests each idea with users before sketching ☐ It records every idea as an annotated sketch

### 4 Key terms: write the letter of the matching description next to each term.

| | | |
|---|---|---|
| 1. **Morphological analysis** | ____ | **A** Collage of images on a mood/theme to inspire ideas and set an aesthetic direction. |
| 2. **Brainstorming rule** | ____ | **B** Taking an idea from one context and applying it in another (e.g. F1 suspension → trainer cushioning). |
| 3. **Analogy** | ____ | **C** A matrix of design variables and options; combine one option from each to generate many ideas. |
| 4. **Lifestyle board** | ____ | **D** Record every idea and don't criticise; build on each other's ideas within a time limit. |
| 5. **Mood board** | ____ | **E** Images showing the lifestyle of the typical user, to understand the target market. |

**5 Explain what is meant by ‘Synthesis’.** (2 marks)

**6 Explain what is meant by ‘Refinement’.** (2 marks)

**Check and reflect**
Mark your answers with the answer sheet. Go back to the checklist on the site and re-rate each “I can” statement above. Which one still needs work?